

# Mengyi Chen

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## EDUCATION

### National University of Singapore

*Ph.D. in Mathematics (Expected Aug. 2027)*

Singapore

*Aug. 2023 – Present*

- Supervisor: Qianxiao Li
- Research Interests: Machine learning for modeling the macroscopic dynamics of complex systems; multiscale modeling; AI for materials

### Columbia University

*Visiting Student, Department of Applied Physics and Applied Mathematics*

New York, USA

*Jan. 2022 – May 2022*

### University of Chinese Academy of Sciences

*B.Sc. in Mathematics and Applied Mathematics*

Beijing, China

*Sep. 2019 – Jun. 2023*

## PUBLICATIONS

- [1] **Mengyi Chen**, Peichen Zhong, Zihan Zhang, Qianxiao Li. “Learning Ab Initio Phase-Field Models.” *arXiv preprint arXiv:2610.01432*, 2026.
- [2] Zihan Zhang\*, **Mengyi Chen\***, Qianxiao Li, Peichen Zhong. “Interfacial Melting as a Thermodynamic Indicator of Solid-State Synthesizability.” *arXiv preprint arXiv:2606.22885*, 2026.  
\* *These authors contributed equally.*
- [3] Zhichao Han, **Mengyi Chen**, Qianxiao Li. “Learning Permutation-Invariant Macroscopic Dynamics.” *International Conference on Machine Learning (ICML)*, 2026 (accepted).
- [4] **Mengyi Chen**, Pengru Huang, Kostya S. Novoselov, Qianxiao Li. “Scalable Learning of Macroscopic Stochastic Dynamics.” *Physical Review Materials*, 10(3):033805, 2026.
- [5] **Mengyi Chen**, Qianxiao Li. “Learning Macroscopic Dynamics from Partial Microscopic Observations.” *Advances in Neural Information Processing Systems (NeurIPS)*, 37:48996–49021, 2024.
- [6] Sheng Gong, Yumin Zhang, Zhenliang Mu, Zhichen Pu, Hongyi Wang, Xu Han, Zhiao Yu, **Mengyi Chen**, Tianze Zheng, Zhi Wang, Lifei Chen, Zhenze Yang, Xiaojie Wu, Shaochen Shi, Weihao Gao, Wen Yan, Liang Xiang. “A Predictive Machine Learning Force-Field Framework for Liquid Electrolyte Development.” *Nature Machine Intelligence*, 7(4):543–552, 2025.

## WORK EXPERIENCE

### ByteDance AML Research Intern

Dec. 2022 – Jul. 2023

*Beijing, China*

- Contributed to developing equivariant graph neural network models for machine learning force fields in the ByteDance AI Molecular Simulation Booster (BAMBOO) project

## HONORS & AWARDS

|                                                        |      |
|--------------------------------------------------------|------|
| NUS Faculty of Science PhD Conference Award            | 2026 |
| NUS Graduate Tutor Merit Award (Semester 2, AY2025/26) | 2026 |
| NeurIPS 2024 Travel Award                              | 2024 |
| NUS Faculty of Science Dean’s Graduate Fellowship      | 2023 |
| UCAS Overseas Graduate Studies Fellowship (Top 5%)     | 2023 |
| UCAS Second-Class Scholarship                          | 2022 |

## PROFESSIONAL SERVICE

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|-----------------------------------------------------------------------------------------------------|------|
| Co-organizer, <i>ICLR 2025 Workshop on Machine Learning Multiscale Processes (MLMP)</i> , Singapore | 2025 |
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## SKILLS & INTERESTS

**Languages:** Chinese (native), English (fluent)

**Interests:** Rock climbing, violin